

EMT

1. Write down Maxwell's wave equations for free space with their physical significance. Hence prove that velocity of EM wave is equal to the velocity of light. (5)
2. Write down the Maxwell's modification of Ampere's law. (2)
3. What is the difference between conduction current and displacement current? Write down the significance of displacement current. (5)
- 3a. Displacement current arises due to _____
4. An electric wave is represented by $E = E_0 \cos(1000x - 5000t)$. Calculate the wavelength and speed of propagation of the wave. Show that the displacement current is negligible compared to the conduction current at any frequency when the conductor is subjected to an alternating field. (2+2)
5. A circular conducting disc of 5 cm radius maintains an angular motion of 1000 rpm with its plane perpendicular to the magnetic field. If due to this motion, the induced e.m.f. between the centre and the edge of the disc is 3.22 mV, calculate the magnetic flux density. (4)
6. A conductor of length L placed in a magnetic field of magnetic flux density $\vec{B}$. If the current I is flowing through the conductor, then how much force is experienced by it? (1)
7. What are dielectrics? Differentiate between polar and nonpolar dielectrics with examples. (4)
8. Give one examples of solid, liquid and gaseous dielectrics. (3)
9. Give three applications of dielectrics. (3)
10. What is Dielectric constant? What is its value in vacuum and conductor. (2+1+1)
- 10a. What do you mean by the 'dielectric constant of glass is 7.0'
11. A capacitor is made up of a dielectric material of $\epsilon_r = 4.2$. It has an effective surface area of 0.09 m^2 and capacitance of $10 \text{ } \mu\text{F}$. Calculate the field strength if a potential difference of 5V exists across the capacitor. Given $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$ (5)
12. Write the SI unit and dimensions of Polarizability. (2)
13. Establish the relation between $\vec{D} = \epsilon_0 \vec{E} + \vec{P}$, where the parameters have their usual meaning. (3)
14. A potential of 10V is applied across a parallel plate capacitor having area of $6.45 \times 10^{-4} \text{ m}^2$ and plate separation is 0.002m. If a material having dielectric constant of 6.0 is placed within the region between the plates calculate the amount of charge stored in the plate and net amount of polarization.

15. Find the relation between electronic polarizability and atomic radius. (4)
16. Define magnetization. Obtain a relation between magnetic permeability and magnetic susceptibility. [5]
17. Define magnetic susceptibility and permeability. Obtain the relation between them. (2+3)
18. The value of permeability of a material is 0.12 N/A^2 . What is relative permeability and magnetic susceptibility?
19. Compare diamagnetic, paramagnetic and ferromagnetic materials. (6)
20. Give the examples of diamagnetic, paramagnetic and ferromagnetic materials. (3)
21. For which type of materials magnetic susceptibility is negative?
22. Define magnetic domain of ferromagnetic materials. (2)
23. State and explain Curie's law for ferromagnetism
24. Differentiate between soft and hard magnetic materials (3)
25. What is hysteresis? Draw a hysteresis curve for a sample iron. Explain the curve step by step.